

ENV 6106

Air Pollution Modeling

Spring 2023

Topics

- Introduction to air pollution
- Expectation
 - Know CAA, CAP & NAAQS, HAP & NESHAP, SIP
 - Understand how CAPs and HAPs are regulated
 - Understand emission types and sources of CAPs
- Readings
 - Cooper and Alley: Chapter 1.1, 1.2, 1.4, 1.5, 1.6, 1.8, 1.9

Regulation

Brief U.S. Federal Legislative History

- Mid 1800s: industrialized cities were well established as very dirty places
 - New York, Chicago, Pittsburgh etc...



Pittsburgh from the Salt Works at Saw Mill, by William T. Russell, 1843
(<http://explorepahistory.com/displayimage.php?imgId=1-2-D44>)

Brief U.S. Federal Legislative History

- By turn of the century, air pollution becoming recognized as a local health problem
- 1907: The International Association for the Prevention of Smoke established
 - Later the Air Pollution Control Association, now the Air & Waste Management Association
- Automobile revolution and beginnings of American consumer culture

Brief U.S. Federal Legislative History

- 1940s: LA haze; Donora (PA) smog (20 deaths)
- 1952: London smog (4000 deaths)
- 1955: Clean Air Legislation (1st Fed. regulation)
 - Authorized Public Health Service (Dept. of Health, Education, and Welfare: DHEW) to:
 - Conduct air pollution research and training programs
 - Provide technical assistance to state and local governments

London smog
(www.klimaforschung.net)



Donora, PA at noon on
October 29, 1948
<http://pabook.libraries.psu.edu/palitmap/DonoraSmog.html>

Brief U.S. Federal Legislative History

- 1960s: Social change and significant national environmental movements
 - Sierra Club, Wilderness Society
 - 1962: Rachel Carson's 'Silent Spring'
- 1963: Clean Air Act (CAA)
 - Broadened federal role in air quality management
 - Grants for program development/improvement of state and local air pollution control
 - Research, training, and technical assistance
 - Federal enforcement authority for interstate air pollution problems
 - Federal research responsibility for automobile and SO_x pollution
 - Authority to compile and publish criteria for pollutants

Brief U.S. Federal Legislative History

- 1965: Motor Vehicle Air Pollution Control Act
 - Amendment to 1963 CAA
 - Established the National Air Pollution Control Administration (NAPCA)
 - Authorized DHEW/NAPCA to promulgate motor vehicle emissions standards

Brief U.S. Federal Legislative History

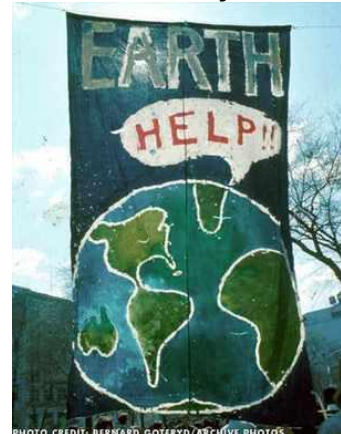
- 1967: Air Quality Act (Amendment to CAA)
 - Institution of a regional approach to air pollution control and standard setting
 - Air Quality Control Regions (AQCR)
 - NAPCA to develop/issue air quality criteria and control technique info
 - States to develop state air quality standards, state implementation plan (SIP)



Brief U.S. Federal Legislative History

- 1970: USEPA established (NAPCA eliminated)
- 1970: Clean Air Act Amendments
 - Provided for federal enforcement authority
 - Stringent auto emissions and fuel standards
 - National Ambient Air Quality Standards (NAAQS)
 - Six Criteria Air Pollutants (CAP)
 - New Source Performance Standard (NSPS)
 - National Emissions Standards for Hazardous Air Pollutants (NESHAPS)

First earth day, 1970



Brief U.S. Federal Legislative History

- 1977: Clean Air Act Amendments
 - Required review of NAAQS every 5 years
 - Non-compliance penalties
 - Prevention of Significant Deterioration (PSD)
 - New Source Review (NSR) for major sources
 - Grant USEPA authority to regulate ozone destroying chemicals
- 1977: National Atmospheric Deposition Program (NADP) established
 - Monitoring trends of acidity in precipitation

Brief U.S. Federal Legislative History

- 1978: US ban manufacture and sale of CFCs for spray cans
- 1980: Acid Precipitation Act
 - National Acid Precipitation Assessment Program (NAPA)
 - Enlarged NADP to National Trends Network (NTN)
- 1980: Election of Ronald Reagan
 - “Onslaught” on environmental regulations

Brief U.S. Federal Legislative History

- 1990: Clean Air Act Amendments
 - Reauthorization and update of CAA
 - Strengthen measures for attaining air quality standards
 - Boundaries of non-attainment areas to include all areas in metropolitan area
 - Expanded regulation of HAPs
 - Max achievable control technology (MACT)
 - Establish program for prevention of accidental releases
 - Updated authority and specific requirements for stratospheric ozone depletion
 - Expanded enforcement powers and penalties
 - Criminal sanctions for negligent violations

Brief U.S. Federal Legislative History

- 2005: Clean Air Interstate Rule (CAIR)
 - Cap and trade for O₃ and PM
 - Replaced by Cross-State Air Pollution Rule (CSAPR) in 2011
- 2007: Supreme court ruling providing EPA authority over CO₂
- 2015: Volkswagen emissions scandal (\$2.8B fine)
- 2015: Clean Power Plan: reduce carbon emissions from power plants
- 2016: Mr. Donald Trump elected POTUS
 - Many environmental policy changes followed
 - Partial list of rollbacks by recent admin ([Link](#)) & ([Link](#))

Brief U.S. Federal Legislative History

- 2017: Dismantle of Clean Power Plan
 - Held by supreme court in 2016
 - Replaced with Affordable Clean Energy rule in 2019 ([Link](#))
 - One example of Clean Coal: Kemper Project ([Link](#))
- 2017 & Beyond: ??????
 - Pull out of Paris climate accord
 - Won't allow CA set its own rules on air quality
 - Allow oil & gas drilling in all US water
 - Relax fuel economy rules
 - Inflation Reduction Act

More on CAPs and HAPs

Emission sources

- Primary vs secondary
 - Directly emitted vs formed in the air
- Natural vs man-made (anthropogenic)
- Stationary vs mobile
 - Stationary stacks, moving cars etc...
- And also by shapes
 - Point, nonpoint, area, volume, line etc....

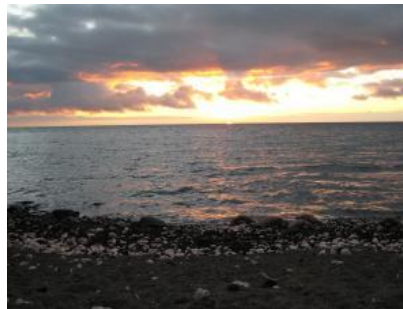
Natural Sources

Volcano



USGS image by C. Newhall - September, 1984 (www.geology.com)

Sea



Credit: Photo by Michele Hogan
<http://www.sciencedaily.com/releases/2007/09/070919175542.htm>

Forest



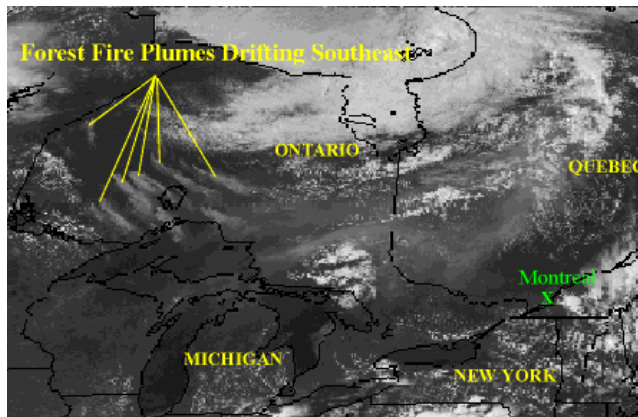
www.ivm.vu.nl/en/projects/Departments/economics/Reduced_Emission_from_Deforestation_and_Degradation_in_the_Southern_Cardamom_Ecosystem_Cambodia/index.asp Forest

Dust storm



NOAA Photo Library, Historic NWS collection

Forest fire



Satellite image for June 14, 1996, at 1545 EST
(www.radar.mcgill.ca/forest_fire.html)

Hot springs



www.phase2.net/claygraphics/Images/bruce%20images/Tahcharian%20for%20Springs.jpg

Anthropogenic Sources

Power plants



(David Klobucar/Associated Press)
<http://www.cbc.ca/news/canadavotes/story/2008/09/19/f-carbontaxprimer.html>

Automobile



www.hybridmile.com/news/obama-supports-californi/

Industry



pollutionarticles.blogspot.com/2009/02/carbon-emissions-causing-huge-ocean.html

Lawn care



www.re-nest.com/uimages/re-nest/mowerwoody052509.jpg

Farms



www.extension.iastate.edu/Pages/communications/EPC/Su02/emissions.html

Land fill



Landfill <http://www.jber.af.mil/shared/media/photodb/photos/2012/08/120821-F-BP133-050.jpg>

CAPs & HAPs

- 6 Criteria Air Pollutants (CAPs):
 - O₃, CO, NO₂, SO₂, Pb, PM (PM_{2.5} & PM₁₀)
 - Concentration level set in NAAQS (in CAA)
 - Ubiquitous, most harmful to health and environment
- 187 Hazardous Air Pollutants (HAPs)
 - A.K.A. toxic air pollutants or air toxics
 - Known/suspected to cause cancer, or other serious health or environmental effects
 - Goal: reduce emissions (NESHAPS)
 - Primary, man-made emissions dominates

Particulate Matter (PM)

- A.K.A. aerosol, can be a solid/liquid/mixed
- PM_{10} & $PM_{2.5}$
 - PMs with aerodynamic diameter < 10 or $2.5 \mu\text{m}$
 - $1 \mu\text{m} = 1 \times 10^{-6} \text{ m}$
 - Can lodge in airway once inhaled.
 - $PM_{2.5}$ penetrate deep into lungs (alveoli region)
- Many emission sources
 - Fire, dust, agriculture, fuel combustion, cars etc...
 - Both primary/secondary, natural/man-made sources
 - Secondary organic aerosol (SOA) is a major research topic

Nitrogen Dioxide (NO₂)

- Usually NO and NO₂ are measured together
 - NO eventually oxidized to NO₂
 - NO + NO₂ = NO_x (different from NO_y and NO_z!)
- Mostly man-made, primary emissions (fuel combustion at high temperatures)
 - Mobile source, power plants dominates
- Some natural sources
 - Lightning generate NO_x
 - Some plants also emit NO_x

Ground-level ozone (O_3)

- One major component of photochemical smog
- Ground-level O_3 BAD; stratospheric O_3 GOOD.
 - Health effect when exposed near ground
 - Stratospheric O_3 layer absorbs harmful UV
- Secondary pollutants, negligible direct emissions
 - Primarily from NO_x and VOC, mixing with sunlight
 - VOC: Volatile Organic Compounds
 - From evaporation of organic solvents, incomplete combustion etc...
 - Trees emits VOC (biogenic): mostly isoprene
 - Globally, trees emits more VOC than man-made sources
 - Ronald Reagan (1981): Trees cause more pollution than automobiles do

Sulfur Dioxide (SO₂)

- From combustion of sulfur-containing fuel
- Primary, anthropogenic sources dominate
 - Coal fire power plants is the biggest source
 - Also industry, mobile, fire (wild + prescribed)
 - Some natural source: sulfur springs
- Major contributor to acid rain, especially northeastern US

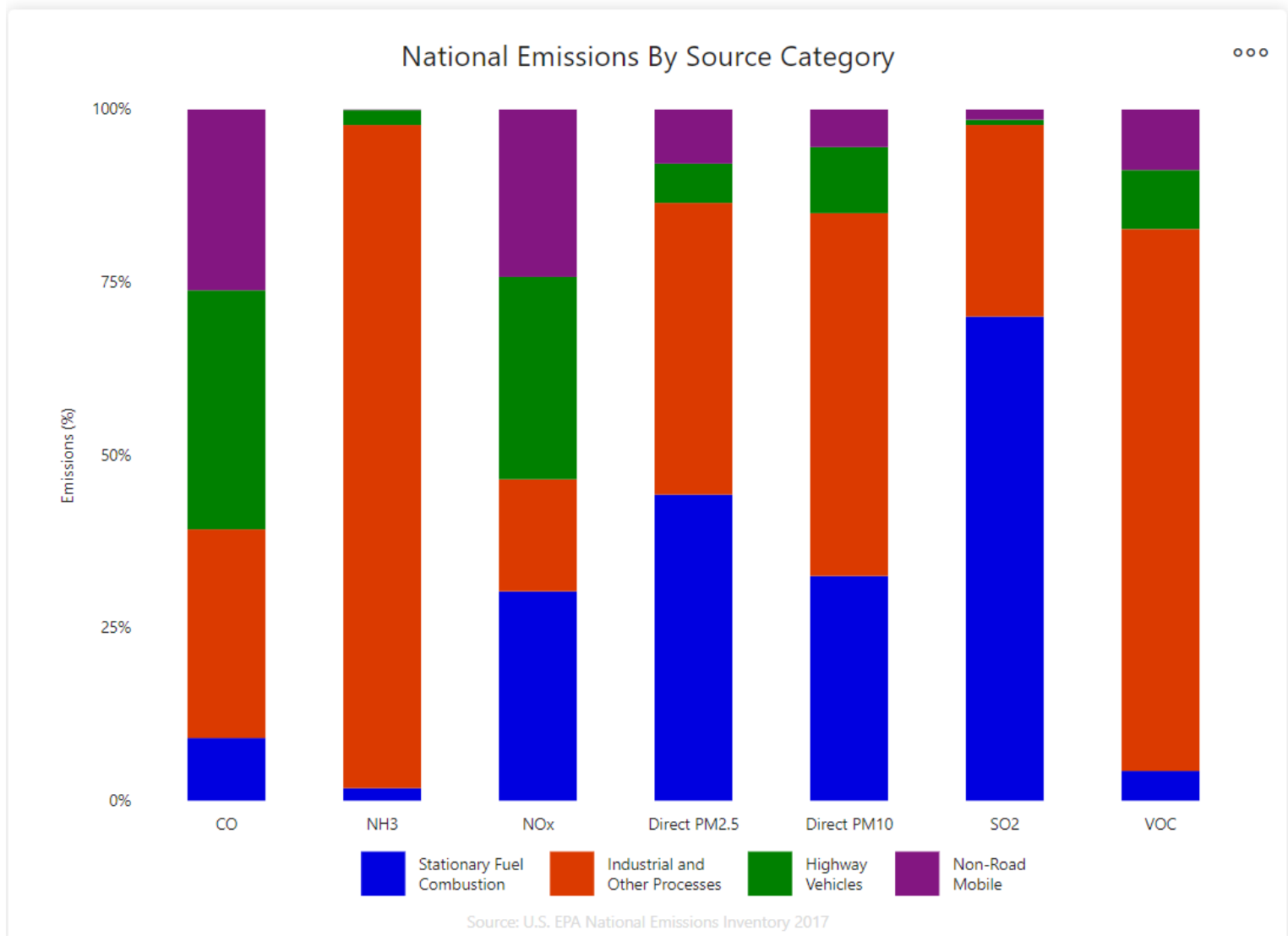
Carbon Monoxide (CO)

- Colorless, odorless, and tasteless gas
- Mostly primary, from incomplete fuel combustion
 - Mobile source, fire are the biggest ones
- Some plants emits CO (biogenic)
- CO binds with hemoglobin, reduce oxygen carrying capacity of blood
- Don't BBQ indoor

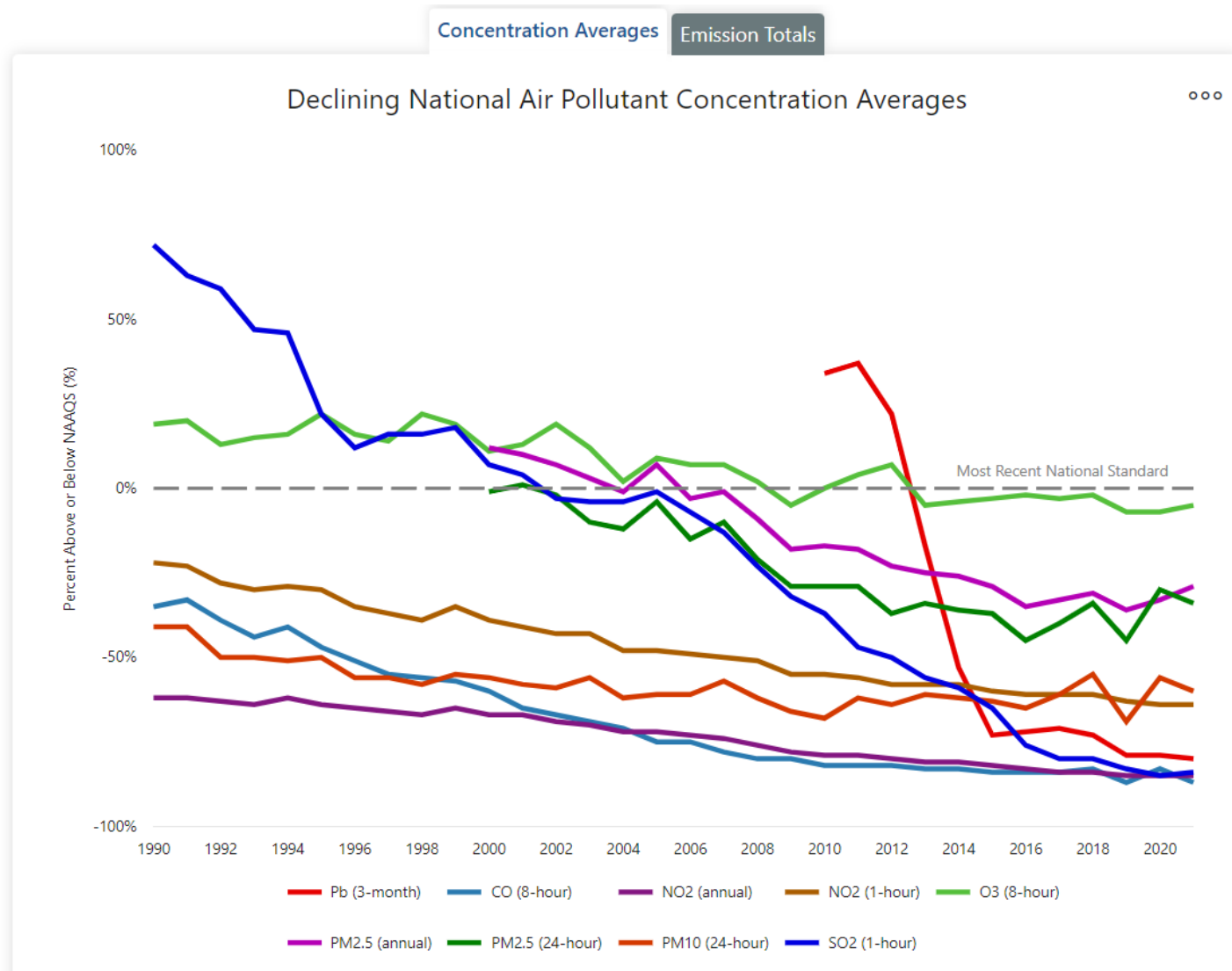
Lead (Pb)

- Particulate lead
- Combustion of fuels with lead additives
- Leaded gas phased out → lead in ambient air decreased by 99%
 - Leaded gas invented by Thomas Midgley Jr. who also brought us Freon (CFC)
- Lead particularly toxic to children
- Houses built before 1978 has lead paint

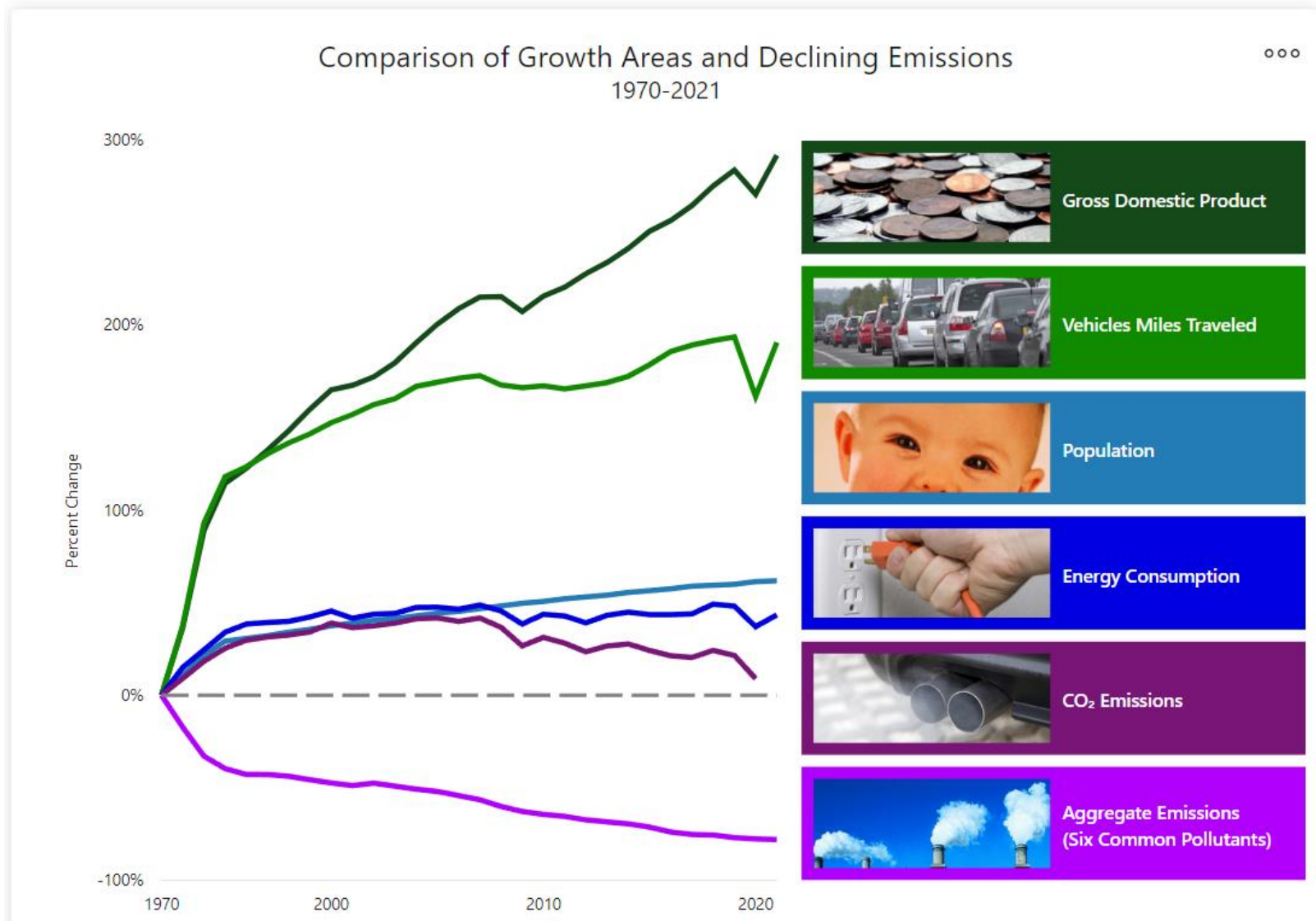
Emission sources



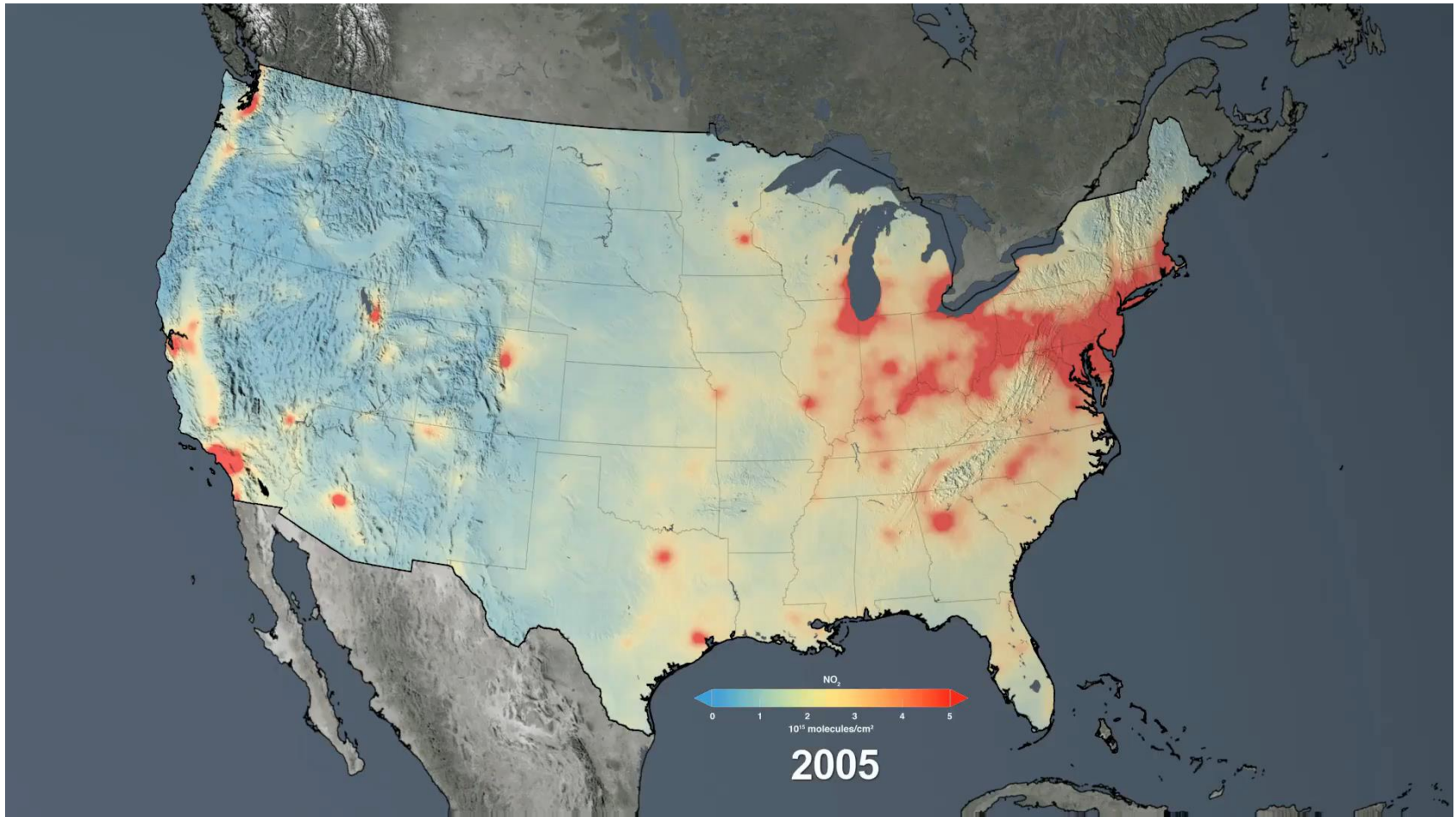
Concentration trend in the US



Emission trend in the US



NASA satellite NO₂ concentration



More information on trend of CAPs see:

https://gispub.epa.gov/air/trendsreport/2018/#naaqs_trends

Also see here: <https://airquality.gsfc.nasa.gov/>

NASA satellite SO₂ concentration

